CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Technology (B.Tech)

University Examination December 2010

Semester – I (CL/EC/ME) (IT/CE/EE)

Engineering Physics PY - 101

Date: 11.12.10, Saturday **Time: 10:30 am to 01:30 pm** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION - I

Q. 1	(a)	Derive and explain the ratio of probabilities of stimulated emission and absorption in terms of Einstein's coefficients.	3
	(b)	What are ultrasonic waves? Explain any three applications of ultrasound.	4
Q. 2	(a)	You are given a ruler, vernier caliper and a micrometer screw gauge to measure the diameter of a coin. Which of the three instruments given, do you think to make the most precise measurement? Why? List and discuss the different types of errors that may present in your measurements.	4
	(b)	(i) What is Weber- Fechner law? Derive the equation for sensitiveness for ear.	3
		(ii) Define threshold of feeling and dead effect.	2
	(c)	Describe different types of fibers by giving the refractive index profile and propagation details.	5
		OR	
Q. 2	(a)	(i) In successive measurements for the period of oscillation of a simple pendulum, the readings turn out to be 2.63 s, 2.56 s, 2.42 s, 2.71s and 2.80 s. Calculate the absolute error, relative error and percentage error.	2
		(ii) The Power P = VI, where V = (500 ± 0.5)V and I = (50 ± 0.2)A. Find the percentage error in P.	2
	(b)	Explain any two methods of detection of ultrasonic sound. Ultrasonic source of 0.09MHz sends down a pulse towards the seabed which returns after 0.55s. The velocity of sound in sea water is 1700m/s. Calculate the depth of sea and wavelength of pulse.	5
	(c)	Distinguish between holography and ordinary photography. How does one construct and reconstruct hologram?	5
Q.3	(a)	Define absorption coefficient. Discuss the factors affecting the acoustics of buildings and give their remedies.	7

	(b)	Explain the terms: (i) absorption, (ii) spontaneous emission, (iii) stimulated emission, (iv) pumping mechanism, (v) population inversion, (vi) optical cavity and (vii) metastable state	7

OR

Q.3	(a)	(i) Define pitch, loudness, timbre and intensity.	4
		(ii) Calculate the intensity level in dB at a distance of 15m, from a source which radiates energy at the rate of 3.56W. The reference intensity is $100 W/m^2$	3
	(b)	What are the characteristics properties of a laser? With the help of suitable diagrams, explain the principle, construction, and working of a CO_2 laser.	7

SECTION - II

Q.4	(a)	A superconducting material has a critical temperature of 3.7 K and a magnetic field of 0.0306 T at 0K. Find the critical field at 2 K.	3
	(b)	Define the terms atomic radius and atomic packing factor. Calculate the same for BCC structure.	4
Q.5	(a)	Explain Hall effect. Show that for an n-type semiconductor the hall coefficient $R_H = -1/ne$.	7
	(b)	Why nanomaterials exhibit different properties? Explain with suitable examples.	7

OR

Q.5	(a)	Describe Josephson effect and their applications. A Josephson junction having a voltage of 8.5 μV across its terminals, then calculate the frequency of the alternating current. [$h= 6.626 \times 10^{-34}$ Js]	7
	(b)	Write a note on nanoscience and nanotechnology. Write any five applications of nanomaterials.	7
Q.6	(a)	What are the characteristics of X-rays? How are X-rays generated?	7
	(b)	What do you understand by Miller indices of a crystal plane? Show that in a cubic crystal the spacing between consecutive parallel planes of Miller indices (hkl) is given by $d = \frac{a}{\sqrt{h^2+k^2+l^2}}$	7

OR

Q.6	(a)	(i) The distance between (100) planes in a body centered cubic structure is 0.203nm. What is the volume of the unit cell? What is the radius of the atom?	4
		(ii) Sketch the following planes of a cubic cell: (020), (130) and (101)	3
	(b)	(i) What are superconductors? Describe the effect of magnetic field, current, pressure and temperature.	5
		(ii) Mention any two industrial applications of superconductors.	2